

Summer Term 2026

Investment and Financial Management

Sessions 11+12: Capital structure and taxes

Berk/DeMarzo (2024): Corporate Finance, 6th edition, Pearson: Chapter 14 and 15)

-
- 1. The interest tax deduction**
 - 2. Valuing the tax shield**
 - 3. Optimal capital structure**
-

1. The interest tax deduction

2. Valuing the tax shield

3. Optimal capital structure

■ Intuition of interest tax deduction

Corporations pay taxes on their profits after interest payments are deducted. Thus, interest expense reduces the amount of corporate taxes.

■ German example of interest tax deduction

- The marginal tax rate on equity income (assuming full distribution) is about 48% vs. 30% for debt income.
- Hence, in order to generate the same after tax income, in case of equity financing the before tax income must be higher by a factor of $(1 - 0.3) / (1 - 0.48) = 1.35$.

➡ This creates a strong incentive to use debt instead of equity

▪ Example of interest tax deduction

Problem

- Consider Macy's which had earnings before interest and taxes (EBIT) of approximately \$2.8 billion in 2014. Its marginal corporate tax rate was 35%.
- Comparison of net income with and without leverage:
 - **With leverage:** Interest expenses of about \$400 million
 - **Without leverage:** No interest expenses and thus a higher net income

	With Leverage	Without Leverage
EBIT	\$2800	\$2800
Interest expense	-400	0
Income before tax	2400	2800
Taxes (35%)	-840	-980
Net income	\$1560	\$1820

▪ Example of interest tax deduction

Problem

- Comparison of capital structure with and without leverage:
 - **With leverage:** Macy's debt obligations reduced the value of its equity. However, Macy's was able to pay out \$1,960 million in total to its investors.
 - **Without leverage:** The total amount available to all investors was higher with leverage. Macy's was able to pay out \$1,820 million in total to its investors.

	With Leverage	Without Leverage
Interest paid to debt holders	400	0
Income available to equity holders	1560	1820
Total available to all investors	\$1960	\$1820

■ Definition of the interest tax shield

The interest tax shield is the reduction in taxes paid due to the tax deductibility of interest.

■ Formula of the interest tax shield

$$\text{Interest tax shield} = \text{Corporate tax rate} \times \text{Interest payments}$$

Problem

- The interest tax shield of Macy's
 - The gain is equal to the reduction in taxes with leverage: \$980 million – \$840 million = \$140 million.
 - The interest payments provided a tax savings of $35\% \times \$400 \text{ million} = \140 million .

■ Calculation of the interest tax shield

Problem

Shown below is the pro-forma forecasted income statement for D.F Builders (DFB), Given its marginal corporate tax rate of 25%, what is the amount of the interest tax shield for DF Bin year 2023 through 2026?

DFB Income statement (\$ million)	2023	2024	2025	2026
Total sales	\$3369	\$3706	\$4077	\$4432
Cost of sales	-2359	-2584	-2867	-3116
Selling, general, and administrative expense	-22	-25	-27	-29
Depreciation	-226	-248	-276	-299
Operating income	762	849	907	988
Other income	7	8	10	12
EBIT	769	857	917	1000
Interest expense	-50	-80	-100	-100
Income before tax	719	777	817	900
Taxes (25%)	-180	-194	-204	-225
Net income	\$539	\$583	\$613	\$675

▪ Calculation of the interest tax shield

Solution

From Eq.15.1, the interest tax shield is the tax rate of 25% multiplied by the interest payments in each year:

(\$ million)	2023	2024	2025	2026
Interest expense	-50	-80	-100	-100
Interest tax shield (25% × interest expense)	12.5	20	25	25

Thus, the interest tax shield enabled DFB to pay an additional \$82.5 million to its investors over this period.

1. The interest tax deduction

2. Valuing the tax shield

3. Optimal capital structure

▪ Valuation of the interest tax shield

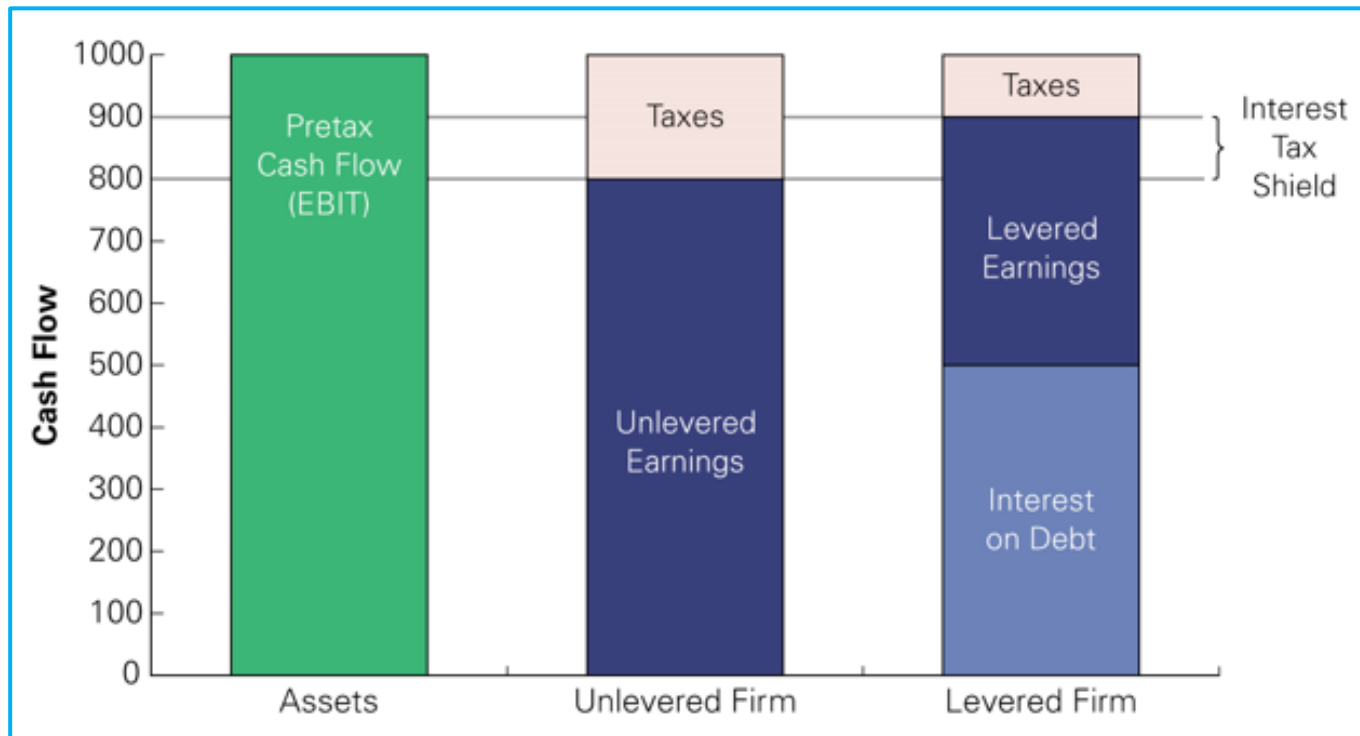
- When a firm uses debt, the interest tax shield provides a corporate tax benefit each year.
- The cash flows a levered firm pays to investors will be higher than they would be without leverage by the amount of the interest tax shield:

$$\left(\begin{array}{c} \text{Cash flows to} \\ \text{investors with leverage} \end{array} \right) = \left(\begin{array}{c} \text{Cash flows to} \\ \text{investors without leverage} \end{array} \right) + (\text{Interest tax shield})$$

- This benefit is calculated as the present value of the firm's expected future interest tax savings.

2. Valuing the tax shield

- Valuation of the interest tax shield: Unlevered versus levered firm



MM proposition I (with taxes):

The total value of the levered firm exceeds the value of the firm without leverage due to the present value of the tax savings from debt.

$$V^L = V^U + PV(\text{Interest tax shield})$$

V^L = Market value of a levered firm

V^U = Market value of an unlevered firm

$PV(\text{Interest tax shield})$ = Present value of the interest tax shield

▪ Valuation of the interest tax shield without risk

Problem

Suppose DFB restructures its existing debt, and will instead pay \$80 million in interest each year for the next 10 years, and then repay the principal of \$1.6 billion in year 10. These payments are risk free, and DFB's marginal tax rate will remain 25% throughout this period. If the risk-free interest rate is 5%, by how much does the interest tax shield increase the value of DFB?

- **Valuation of the interest tax shield without risk**

Solution

In this case, the interest tax shield is $25\% \times \$80 \text{ million} = \20 million each year for the next 10 years. Therefore, we can value it as a 10-year annuity. Because the tax savings are known and not risky, we can discount them at the 5% risk-free rate:

$$\begin{aligned} \text{PV(Interest Tax Shield)} &= \$20 \text{ million} \times \frac{1}{0.05} \left(1 - \frac{1}{1.05^{10}} \right) \\ &= \$154 \text{ million} \end{aligned}$$

The final repayment of principle in year 10 is not deductible, so it does not contribute to the tax shield.

▪ The interest tax shield with permanent debt

Typically, the level of future interest payments is uncertain due to changes in:

- marginal tax rate,
- amount of debt outstanding,
- interest rate on that debt,
- risk of the firm

For simplicity, we will consider the special case in which the above variables are kept constant.

- **The interest tax shield with permanent debt**

Suppose a firm borrows debt D and keeps the debt permanently. If the firm's marginal tax rate is τ_c , and if the debt is riskless with a risk-free interest rate r_f , then the interest tax shield each year is $\tau_c \times r_f \times D$, and the tax shield can be valued as a perpetuity.

$$\text{PV(Interest tax shield)} = \frac{\tau_c \times \text{Interest}}{r_f} = \frac{\tau_c \times (r_f \times D)}{r_f} = \tau_c \times D$$

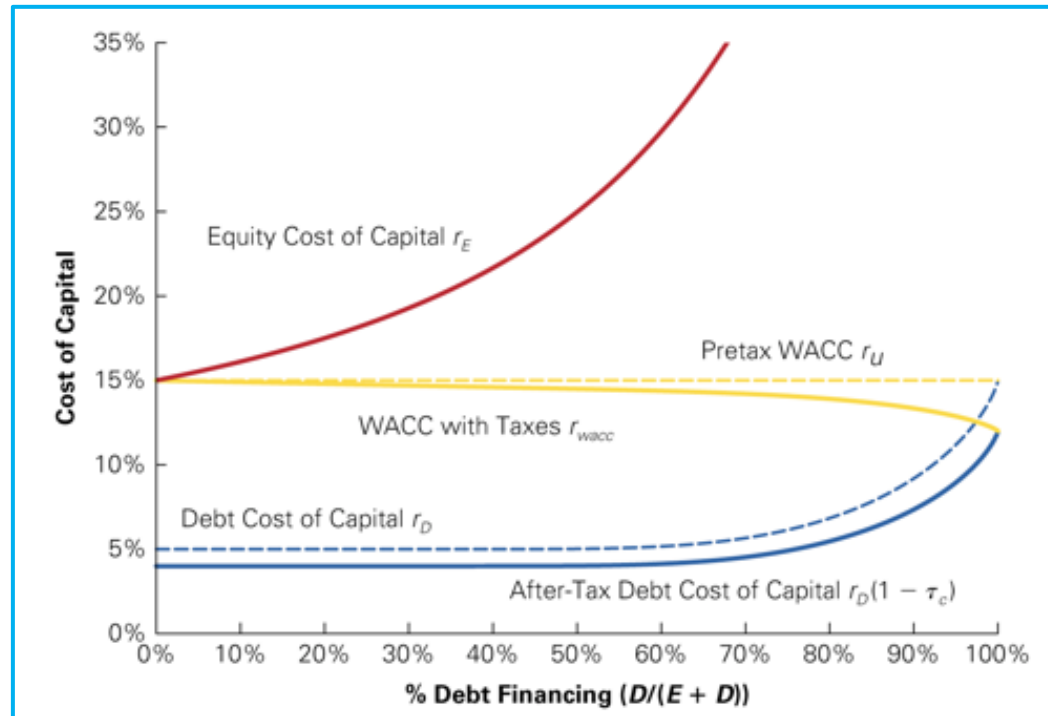
- **The WACC with taxes**

With tax-deductible interest, the effective after-tax borrowing rate is $r_D(1 - \tau_c)$ and the weighted average cost of capital becomes:

$$r_{WACC} = \frac{E}{D + E} r_E + \frac{D}{D + E} r_D (1 - \tau_c)$$
$$r_{WACC} = \underbrace{\frac{E}{D + E} r_E + \frac{D}{D + E} r_D}_{\text{Pre-tax WACC}} - \underbrace{\frac{D}{D + E} r_D \tau_c}_{\text{Reduction due to interest tax shield}}$$

2. Valuing the tax shield

- **Pre-tax WACC = The firm's unlevered cost of capital**
 - Is constant, reflecting the required return of the firm's investors based on the risk of the firm's assets
- **Effective after-tax WACC = The firm's after-tax cost to the firm**
 - Declines with leverage as the interest tax shield grows



- **The interest tax shield with a target debt-to-equity ratio**
 - When a firm adjusts its leverage to maintain a target debt-to-equity ratio, we can compute its value with leverage, V^L , by discounting its free cash flow using the weighted average cost of capital.
 - The value of the interest tax shield can be found by comparing the value of the levered firm, V^L , to the unlevered value, V^U , of the free cash flow discounted at the firm's unlevered cost of capital, the pre-tax WACC.

- The interest tax shield with a target debt-to-equity ratio

Valuing the Interest Tax Shield with a Target Debt-Equity Ratio

Problem

Western Lumber Company expects to have free cash flow in the coming year of \$4.25 million, and its free cash flow is expected to grow at a rate of 4% per year thereafter. Western Lumber has an equity cost of capital of 10% and a debt cost of capital of 6%, and it pays a corporate tax rate of 35%. If Western Lumber maintains a debt-equity ratio of 0.50, what is the value of its interest tax shield?

■ The interest tax shield with a target debt-to-equity ratio

Solution

We can estimate the value of Western Lumber's interest tax shield by comparing its value with and without leverage. We compute its unlevered value by discounting its free cash flow at its pretax WACC:

$$\text{Pretax WACC} = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D = \frac{1}{1+0.5}10\% + \frac{0.5}{1+0.5}6\% = 8.67\%$$

Because Western Lumber's free cash flow is expected to grow at a constant rate, we can value it as a constant growth perpetuity:

$$V^U = \frac{4.25}{8.67\% - 4\%} = \$91 \text{ million}$$

To compute Western Lumber's levered value, we calculate its WACC:

$$\begin{aligned}\text{WACC} &= \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - \tau_c) \\ &= \frac{1}{1+0.5}10\% + \frac{0.5}{1+0.5}6\%(1 - 0.35) = 7.97\%\end{aligned}$$

Thus, Western Lumber's value including the interest tax shield is

$$V^L = \frac{4.25}{7.97\% - 4\%} = \$107 \text{ million}$$

The value of the interest tax shield is therefore

$$PV(\text{Interest Tax Shield}) = V^L - V^U = 107 - 91 = \$16 \text{ million}$$

- **Example of recapitalizing to capture the tax shield**

Problem

Assume that Midco Industries wants to boost its stock price. The company currently has 20 million shares outstanding with a market price of \$15 per share and no debt. Midco has had consistently stable earnings and pays a 21% tax rate. Management plans to borrow \$100 million on a permanent basis and they will use the borrowed funds to repurchase outstanding shares.

▪ Example of recapitalizing to capture the tax shield

Solution (I)

- **Without leverage**

- $V^U = 20 \text{ million shares} \times \$15/\text{share} = \$300 \text{ million}$

- **With leverage**

- If Midco borrows \$100 million using permanent debt, the present value of the firm's future tax savings is: $PV(\text{interest tax shield}) = \tau_c \times D = 21\% \times \$100 \text{ million} = \$21 \text{ million}$

- Total value of the levered firm: $V^L = V^U + \tau_c D = \$300 \text{ million} + \$21 \text{ million} = \$321 \text{ million}$

- Total value of debt: $D = \$100 \text{ million}$

- Total value of equity: $E = V^L - D = \$321 \text{ million} - \$100 \text{ million} = \$221 \text{ million}$

▪ Example of recapitalizing to capture the tax shield

Solution (II)

- **Share repurchase**

- Although the value of the shares outstanding drops to \$221 million, shareholders will also receive the \$100 million that Midco will pay out through the share repurchase.
- In total, they will receive the full \$321 million, a gain of \$21 million over the value of their shares without leverage.
- Assume Midco repurchases its shares at the current price of \$15/share. The firm will repurchase 6.67 million shares. $\$100 \text{ million} \div \$15/\text{share} = 6.67 \text{ million shares}$
- It will then have 13.33 million shares outstanding. $20 \text{ million} - 6.67 \text{ million} = 13.33 \text{ million}$

▪ Example of recapitalizing to capture the tax shield

Solution (III)

- **Share repurchase**

- The total value of equity is \$221 million; therefore the new share price is \$17.575/share. $\$221 \text{ million} \div 13.33 \text{ million shares} = \$16.575/\text{share}$
- Shareholders that keep their shares earn a capital gain of \$1.575 per share. $\$16.575/\text{share} - \$15/\text{share} = \$1.575/\text{share}$
- The total gain to shareholders is \$21 million. $\$1.575/\text{share} \times 13.33 \text{ million shares} = \21 million
- If the shares are worth \$16.575/share after the repurchase, why would shareholders tender their shares to Midco at \$15/share?

▪ Example of recapitalizing to capture the tax shield

Solution (IV)

- **No arbitrage pricing**

- Arbitrage opportunity: If investors could buy shares for \$15 immediately before the repurchase and they could sell these shares immediately at a higher price afterward.
- Realistically, the value of the Midco's equity will rise immediately from \$300 million to \$321 million after the repurchase announcement.
- With 20 million shares outstanding, the share price will rise to \$16.05 per share. $\$321 \text{ million} \div 20 \text{ million shares} = \$16.05/\text{share}$
- With a repurchase price of \$16.05, the shareholders who tender their shares and the shareholders who hold their shares both gain \$1.05/share as a result of the transaction. $\$16.05/\text{share} - \$15/\text{share} = \$1.05/\text{share}$

▪ Example of recapitalizing to capture the tax shield

Solution (V)

- **No arbitrage pricing**

- The benefit of the interest tax shield goes to all 20 million of the original shares outstanding for a total benefit of \$21 million. $\$1.05/\text{share} \times 20 \text{ million shares} = \21 million
- **When securities are fairly priced, the original shareholders of a firm capture the full benefit of the interest tax shield from an increase in leverage.**

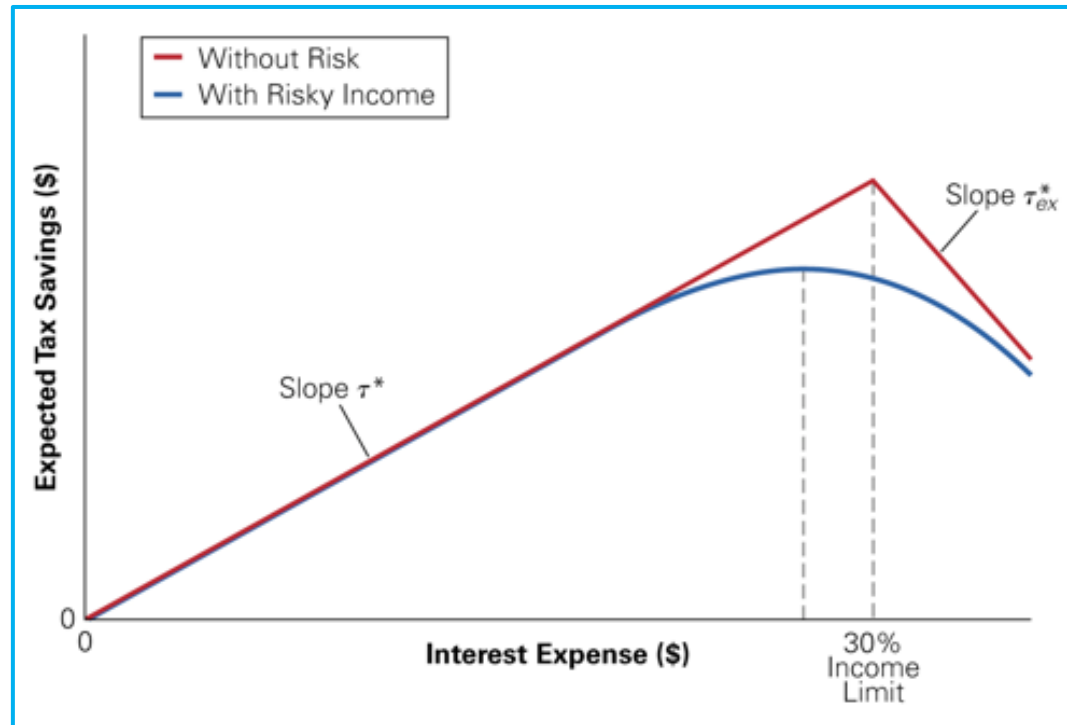
1. The interest tax deduction

2. Valuing the tax shield

3. Optimal capital structure

■ Optimal level of leverage

The optimal level of leverage from a tax saving perspective is the level such that interest equals EBIT. At the optimal level of leverage, the firm shields all of its taxable income and it does not have any tax-disadvantaged excess interest.



- **The Influence of Financial Distress Costs on Capital Structure**
 - Despite the tax shield benefit, hardly any firm is financed 100% with debt
 - This can be explained by the costs of insolvency or financial distress, because the risk of insolvency or financial distress rises with a higher level of leverage
 - In perfect capital markets (MM*), insolvency costs do not exist
 - In case of insolvency outside creditors take over the company management
 - The company continues with its operations or gets liquidated. The liquidation proceeds are equal to the fair value of assets in perfect capital markets

*Modigliani-Miller theorem

▪ **The Influence of Financial Distress Costs on Capital Structure**

- In real world insolvency costs can be enormous. They include:
 - Direct costs (e.g. attorney fees and consulting fees during the insolvency process, ...)
 - Indirect costs (e.g. loss of essential customers, workers and suppliers, panic selling of assets below fair value, lucrative projects can't be implemented during the insolvency phase ...)
- A study estimated the potential costs of financial distress to range between 10% and 20% of the company's value

■ The Static Trade-Off Theory

- There's a Trade-Off between the interest tax shield and financial distress costs
- Each company has an optimal (= minimum WACC) degree of leverage:

$$V^L = V^U + PV(\text{Interest Tax Shield}) - PV(\text{Financial Distress Costs})$$

■ Chapter quiz of session 11+12

1. How do corporate taxes impact a firm value as leverage changes?
2. How does leverage affect a firm weighted average cost of capital?
3. How can shareholders benefit from a leveraged recapitalization when it reduces the total value of equity?
4. What does the trade-off theory predict about a firm's optimal capital structure?